

Budgeted Replanning in History-Dependent Games: Certified Value Frontiers for Revision Calendars

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Abstract

Decision makers seldom receive unlimited opportunities to revise a plan. We study how much revision an adversarial trajectory planner needs when revisions are rationed to a calendar: a public set of epochs at which the player may observe the executed history and seal a new block of commands. The underlying game is history dependent — trailing vessels follow a leader’s executed track, so the current pose does not determine the formation — and the payoff integrates a directional interaction field along the trajectory. We show that the value attainable with at most K revisions, W_K , is computable exactly up to a certified interval; that it is monotone in K by policy inclusion; and that the calendar objective is *not* submodular, so selecting revisions is not a greedy problem — we give a binary-action counterexample. Two certificates make budgets auditable: a constructive deletion certificate that returns a feasible coarse policy and bounds its loss, and an interval-loss certificate for a fully flexible opponent that reduces a budget question to a shortest path over public intervals. On a frozen computational benchmark of three geometries, four speed/distance configurations, two opponent classes and three one-factor ablations, the budget is binding rather than slack: in most cells the smallest budget that attains the fully flexible value is the full budget, and where it is not, the maximizing calendars concentrate on the early engaged epochs. We report the sharpness of each certificate rather than presenting a valid bound as a tight one, and we keep a previously tested hypothesis — that the range-advantaged side values revisions differently — as a falsified result rather than reviving it under a new name.

1 Introduction

1.1 The problem

Decision makers rarely receive unlimited opportunities to revise a plan. A commander who can re-issue manoeuvre orders at every moment is not the same agent as one who may re-issue them twice in an engagement, even when both retain the same ships, the same sensors and the same doctrine. What differs is the *revision calendar*: the set of epochs at which the agent may look at what has actually happened and choose again. Replanning is not free, it cannot be scheduled after the fact, and the value of a revision depends on what the opponent has already done.

This paper studies the resulting optimisation problem in a class of adversarial trajectory games. Two formations move under nonholonomic constraints inside a directional interaction field, and each earns a payoff that integrates the field along the whole trajectory rather than at a terminal instant. The state that matters is a history: trailing vessels follow the leader’s executed track, so the current leader pose does not determine the formation’s shape [7]. Each player is allowed a bounded number of revisions, arranged on a calendar that is fixed and public in advance. Within each interval between consecutive revisions the player seals a block of commands; the opponent sees the block only as it is executed.

The question is deliberately budget-shaped. Given a tolerated loss of performance, how many revisions are needed, when should they occur, and can a computation certify that the chosen calendar is good enough? Three things make this non-trivial. First, the feasible policy set

depends on the calendar, so calendars cannot be compared by evaluating a single pre-computed policy. Second, more revisions need not have diminishing returns: the natural greedy argument fails, because revision opportunities can be complementary. Third, the opponent may itself be revising, and a strategy that is optimal against a committed opponent is not admissible against a flexible one.

1.2 Approach

We treat the calendar as an explicit decision variable and build an exact finite-game formulation around it. Section 2 fixes the history-dependent interaction game, the sealed-block information structure and the budgeted value W_K . Section 3 develops four structural results and the certificates the computation relies on: policy inclusion gives monotonicity in the budget and identifies the endpoints; a counterexample shows that the calendar objective is not submodular, so selection needs more than a greedy argument; a common-update decomposition localises the game at public boundaries; and two certificate constructions bound the loss of a coarsened calendar from above, one by an explicit feasible policy and one by summing losses over public intervals. Section 4 turns these into a solver: exact pricing of revisions against a fixed causal opponent, calendar branch and bound with certified intervals, and reuse of opponent responses across calendars. Section 5 freezes the experimental design, and Section 6 reports it.

1.3 Contributions and scope

1. **A certified budget frontier for history-dependent sealed-block games.** For a fixed opponent class we compute, exactly up to a certified interval, the value attainable with at most K revisions, W_K , over a frozen finite design. The frontier identifies which revisions actually pay and where the budget stops binding.
2. **Structure that survives contact with counterexamples.** We prove budget monotonicity and endpoint equivalence, exhibit a binary-action counterexample showing the calendar objective is not submodular, and prove a common-update decomposition whose scope is stated exactly: a revision by one player alone is not a public boundary while the opponent still holds an unexecuted block.
3. **Two computable loss certificates and an exact pricing recurrence.** A constructive deletion certificate gives a feasible coarse policy and an upper bound on the loss it incurs; an interval-loss bound gives a shortest-path certificate for a budget under a fully flexible opponent. Both are reported with the residual of the computation that produced them.

The scope is bounded and we state it where it binds. The physical design is a *fixed computational benchmark* over three geometries, four speed/distance configurations, two opponent classes and three one-factor ablations; it is not a statistical sample and we do not attach p -values to deterministic solver output. The opponent classes are a committed class and a fully flexible class, and the second certificate applies only to the latter, where public boundaries exist. A previously investigated claim — that the range-advantaged side values revision rights differently from the range-disadvantaged side — was tested at this fidelity and *falsified*; those results stay archived and are disclosed as a negative finding rather than repaired. Our claim is about the frontier and the certificates, not about a new general equilibrium-computation method: the pricing recurrence specialises established extensive-form best-response ideas [1, 4, 6], and completeness of the reported frontier is relative to the frozen design and the certified tolerances.

1.4 Related work

Our setting sits between three literatures. In *dynamic games with information*, an informed player’s value is monotone in the information structure [3]; revisions are not free observations

but binding command blocks, so the inclusion argument applies to policy sets rather than to signal structures. Huang and Zhu study a pursuit–evasion differential game in which observation opportunities are purchased strategically [?]; there the cost is paid for information about the state, whereas here the budget limits when the agent may *act on* information it already has. Chen and Waggoner show that whether additional signals are substitutes or complements depends on the decision problem [2]; our counterexample extends that caution to revision calendars.

In *zero-sum game solving*, double-oracle and column-generation methods converge finitely with response oracles [1, 6], and extensive-form abstractions admit bounds on equilibrium error [4]. We use those tools rather than reinvent them. The distinction is the object being optimised: not a strategy within a fixed information structure, but the information structure itself, subject to an explicit revision budget and with per-calendar certificates.

In *formation control and pursuit–evasion*, leader–follower path-following formations and speed asymmetry under turn constraints are classical. We use a leader–follower line-ahead formation as the numerical instantiation and treat a rigid column as an ablation, so that conclusions are not an artefact of one formation representation.

2 History-dependent interaction and revision rights

2.1 From a directional field to a history payoff

Consider two formations, Blue and Red, moving in the plane over a finite horizon T . Blue maximizes net interaction payoff and Red minimizes it. A leader’s motion is generated by a sequence of bounded turn controls. Trailing vessels occupy fixed arc-length offsets along its executed track; their orientations are the local track tangents. For formation i , vessel k has position and heading

$$x_{ik}(t) = \gamma_i(s_i(t) - \ell_k), \quad \psi_{ik}(t) = \arg \gamma'_i(s_i(t) - \ell_k). \quad (1)$$

The numerical implementation uses a causal polyline and its incoming tangent at an exact station. A straight prehistory supplies the trailing offsets at the initial epoch. Current leader pose alone is generally insufficient to recover the trailing configuration. The analysis therefore retains executed histories rather than identifying states solely by leader position and heading.

Let r_{kl} denote pairwise separation, and $\delta_{kl}^B, \delta_{lk}^R$ the target bearings measured relative to the respective vessel headings. A directional kernel K_i gives the interaction intensity for each pair. The formation-level differential is

$$L(h_t) = \sum_{k,l} \left[K_B(r_{kl}, \delta_{kl}^B, \delta_{lk}^R) - K_R(r_{kl}, \delta_{lk}^R, \delta_{kl}^B) \right]. \quad (2)$$

Here h_t is the complete executed history through time t . The specific kernel used below is a frozen surrogate of an audited game-rule response, not a model calibrated to observed naval engagements. Its firing sectors are piecewise defined; no global differentiability is assumed.

The objective accumulates interaction along the trajectory. The finite benchmark uses the causal endpoint trapezoid

$$J(h_T) = \frac{1}{2} \sum_{t=0}^{T-1} \{L(h_t) + L(h_{t+1})\}. \quad (3)$$

The initial term is a common constant for a fixed instance; every command affects a subsequent state included in the same evaluation horizon. This is a specified discrete objective, not an assertion of exact continuous-time integration. Directional motion can change the ordering of paths because an attractive terminal configuration need not compensate for unfavorable interaction accumulated earlier. We compare the integrated and terminal objectives under the same horizon in a separate ablation.

Let each player have a state-independent finite action menu of size b . Enumerating the b^T complete action paths of each player gives a matrix $A \in \mathbb{R}^{b^T \times b^T}$, where $A(i, j) = J(h_T(i, j))$. This matrix indexes realized physical paths. Its rows are *not* all feedback policies: a policy may select different own paths after different observed opponent prefixes. The distinction is essential when computing adaptation value.

2.2 Public calendars and sealed command blocks

Write $\mathcal{N} = \{1, \dots, T-1\}$. A calendar $S \subseteq \mathcal{N}$ specifies Blue’s revision opportunities; an initial decision at $t = 0$ is compulsory and does not consume the revision budget. Calendars are fixed and publicly known before play. At each $t \in \{0\} \cup S$, Blue observes both executed action prefixes and selects a complete block of commands until its next update. Red follows an independently specified calendar R . The simultaneous current commands and the unexecuted portions of either block are hidden from the opponent. Between updates a player executes its chosen block. At its next update it remembers its earlier choices and the complete executed history.

This is a finite perfect-recall extensive-form game. Revision rights constrain when commands may respond to observations; they do not erase a player’s memory. No noisy observations, damage states, private capability types or history-dependent legal-action restrictions are added to the numerical model.

Let $\mathcal{P}_S$ and $\mathcal{Q}_R$ denote the mixed-policy sets induced by the calendars. Define

$$V(S, R) = \max_{p \in \mathcal{P}_S} \min_{q \in \mathcal{Q}_R} \mathbb{E}_{p,q}[A(i, j)]. \quad (4)$$

Finite minimax supplies a value. We study two fixed opponent conditions: $R = \emptyset$ (committed, denoted C) and $R = \mathcal{N}$ (flexible, denoted F). Results are reported separately. In a calendar comparison, the physical matrix and the opponent’s policy space remain unchanged.

2.3 The budget frontier and the decision target

For $K \in \{0, \dots, T-1\}$, the calendar design problem is

$$W_K(R) = \max_{S \subseteq \mathcal{N}: |S| \leq K} V(S, R). \quad (5)$$

The design variable is a deterministic public calendar. The opponent may respond strategically within its fixed policy class. Hidden randomized calendars and jointly optimized opposing calendars define different design problems and are outside the present formulation.

Let $V_0 = V(\emptyset, R)$, $V_F = V(\mathcal{N}, R)$ and $D = V_F - V_0$. When D is numerically resolved and positive, the smallest budget retaining a fraction $\rho \in (0, 1)$ of full adaptation value is

$$K_\rho = \min\{K : W_K(R) \geq \rho V_F + (1 - \rho)V_0\}. \quad (6)$$

Otherwise we report absolute performance loss. This prevents an almost-zero adaptation premium from producing unstable percentage comparisons.

3 Structure and certificates

3.1 Policy inclusion does not imply diminishing returns

Proposition 1 (Calendar inclusion). *For fixed A and R , $S \subseteq U$ implies $V(S, R) \leq V(U, R)$. Consequently $W_K(R)$ is nondecreasing, with $W_0 = V_0$ and $W_{T-1} = V_F$.*

Proof. An S -policy can be implemented under U by choosing the corresponding parts of each original block and ignoring observations at additional update epochs. The player remembers

Table 1: Main notation and assumptions. All calendar comparisons hold the physical instance and the opponent condition fixed.

Symbol	Meaning
T, b	Horizon and finite per-epoch action-menu size
$A(i, j)$	Accumulated payoff for a complete pair of executed paths
S, R	Public Blue and Red revision calendars; initial decision excluded
$\mathcal{P}_S, \mathcal{Q}_R$	Perfect-recall policies with hidden unexecuted command blocks
$V(S, R), W_K(R)$	Fixed-calendar value and best value with at most K revisions
$[L_S, U_S]$	Recorded finite-game numerical value interval
K_ρ	Minimal budget retaining fraction ρ of full adaptation
Standing assumptions	Known initial state, deterministic causal dynamics, bounded payoff, state-independent legal action menus, full executed-history observation at updates

the original block or the privately sampled pure policy. This embedding is outcome-equivalent against every policy in the same $\mathcal{Q}_R$. Extending the embedding to mixtures and maximizing over the enlarged feasible set proves the value inequality. The budget and endpoint statements follow immediately. $\square$

This is a finite policy-inclusion argument, consistent with established zero-sum information-value results [3]. We use it as a correctness condition, not as an independent novelty claim. It does not compare different physical speeds or different opponent classes.

Proposition 2 (Complementary update opportunities). *Even with binary actions, a committed opponent and a nonnegative bounded payoff, $S \mapsto V(S, C)$ need not be submodular.*

Proof. Take $T = 3$ and payoff $A = \mathbf{1}\{b_1 = r_0\}\mathbf{1}\{b_2 = r_1\}$. The values for $S = \emptyset, \{1\}, \{2\}, \{1, 2\}$ are $1/4, 1/2, 1/2, 1$, respectively. Without updates, independent uniform Blue guesses guarantee $1/4$, and independent uniform Red bits bound the value by $1/4$. With only update 1, Blue matches the observed first bit and guesses the second; with only update 2 it guesses the first and matches the second. Uniform unknown bits bound both values by $1/2$. Both updates permit both matches. Thus the marginal value of update 2 rises from $1/4$ to $1/2$ when update 1 is already present, violating submodularity. $\square$

This example concerns the general finite history-payoff class, not every formation instance. It rules out a generic diminishing-returns justification for greedy calendar selection. The informational-substitutes literature likewise requires substantive conditions on the decision problem [2]. A sufficient special case is a sum of independent reset games with product feasible sets, independent information and no shared opponent resource: product equilibrium strategies give matching lower and upper guarantees for the sum, so each selected update contributes a fixed premium. That special case is modular. The formation history in (1) generally prevents such a reset decomposition.

3.2 Where the game can be decomposed

Theorem 3 (Common-update decomposition). *At a common update $c \in S \cap R$, each complete executed prefix defines a proper subgame. Backward replacement of these subgames by their minimax values preserves the root value. No identification of distinct physical histories is needed.*

Proof. At c both previously selected command blocks have finished. Their executed contents are public, and no private unexecuted command crosses the cut. All histories belonging to either player's information set at or after this prefix remain inside the corresponding subtree. The subtree is therefore a proper subgame. Within it, the maximizing player can guarantee its value and the minimizing player can enforce the same upper bound. Replacing its continuation by that

scalar gives matching guarantees in the preceding game. Induction over common boundaries, in reverse chronological order, proves equality at the root. $\square$

An update of Blue alone is not such a boundary while Red still has an unfinished command block. This restriction separates the result from an incorrect full-information Bellman recursion at every physical epoch.

Corollary 4 (Propagation of subgame error). *If the uniform value error at replacement level l is at most e_l , the root error is at most $\sum_l e_l$.*

Proof. A uniform perturbation of at most e changes the expected payoff of every fixed strategy pair by at most e . Taking the minimum and maximum preserves the bound. Induct over the successive replacements. $\square$

3.3 Exact pricing against a fixed causal opponent

For a fixed opponent policy q , define $C_q(j \mid i)$ as its realization probability of path j when the own path i is externally fixed. This is a counterfactual conditional law; it does not assume that the opponent commits. For committed q , it reduces to q_j . For a mixture of causal pure opponent maps $r_m(i)$ with weights w_m ,

$$C_q(j \mid i) = \sum_m w_m \mathbf{1}\{r_m(i) = j\}. \quad (7)$$

Every positive mixture weight is retained. Initialize $H_T(i, j) = A(i, j)C_q(j \mid i)$. At consecutive own block boundaries $t < e$, apply the recurrence

$$H_t(i_t, j_t) = \max_a \sum_z H_e(i_t a, j_t z), \quad (8)$$

where a is the complete own block and z ranges over the opponent's executed actions during that block. The sum precedes the maximum.

Theorem 5 (Response over the complete permitted policy class). *Recurrence (8) returns the maximum payoff against q over all policies permitted by the own calendar. Its stored maximizing blocks define a feasible pure best response.*

Proof. Causality implies that the counterfactual opponent probability up to time t depends only on the executed prefix, hence is independent of the yet-unchosen own block. For each block, all unobserved opponent continuations must first be summed. At the next own update the continuation can be chosen separately for each newly observed prefix; its optimum has already been computed by the induction hypothesis. The resulting sum followed by a maximum is precisely the feasible decision at t . Backward induction proves root optimality. Prefix probabilities remain unnormalized, so no division by zero is required. The recorded block at each information set uses only its observed prefixes, which proves feasibility, including arbitrary legal choices at zero-weight histories. $\square$

The recurrence takes $O(b^{2T})$ arithmetic and memory for $b \geq 2$, up to a geometric-series factor. Constructing the physical matrix and generating multiple response policies are additional costs. The result specializes established extensive-form best-response ideas; we do not claim a new general equilibrium-computation framework [1, 5].

3.4 Transferable upper bounds and feasible deletion policies

Proposition 6 (Opponent-policy reuse). *For fixed opponent calendar R , a causal opponent policy remains feasible when only the own calendar changes. For any finite pool $Q_0 \subseteq \mathcal{Q}_R$,*

$$V(S, R) \leq \min_{q \in Q_0} \text{BR}_S(q). \quad (9)$$

Proof. The opponent's update times, observed executed prefixes and legal action menu remain unchanged. Its old policy can ignore the new public calendar and use the same map of observations to commands. Its feasibility never depends on observing an unexecuted suffix. Fixing any such policy in the minimax expression gives an upper bound, and the minimum of valid upper bounds is valid. $\square$

To remove updates, we construct a legal coarse policy rather than treating an infeasible fine-calendar strategy as if it remained admissible. Express a full-update source strategy as conditional action kernels. At a coarse update, retain the actual own and opponent prefixes, complete the opponent future by a fixed straight-action suffix, and multiply the source kernels along every possible own block. These products define its block distribution. When a source information set has zero incoming probability, use the uniform legal action distribution. This is an explicit new behavioral policy; no claim is made that it preserves the source's outcome distribution after updates are deleted.

Theorem 7 (Constructive deletion certificate). *Under state-independent legal action menus, the construction gives a feasible policy $\tilde{p}$ for any coarse calendar $S' \subseteq \mathcal{N}$. If U_F bounds V_F from above and $L' = \min_{q \in \mathcal{Q}_R} \mathbb{E}_{\tilde{p}, q} A$, then*

$$0 \leq V_F - V(S', R) \leq U_F - L'. \quad (10)$$

Proof. Each source kernel is normalized. Summing the products over all own blocks therefore gives one by iterated normalization. The construction uses only information observed at the coarse update and a fixed hypothetical suffix. Every block is legal because action menus do not depend on the shadow trajectory. Thus $\tilde{p} \in \mathcal{P}_{S'}$ and $L' \leq V(S', R)$. Combine this inequality with $V_F \leq U_F$ and Proposition 1. $\square$

The certificate can be loose; its usefulness must be measured. It is a full-response bound on an explicit policy, not a generic smoothness bound on the directional field. Existing abstraction theory also distinguishes solution error from the size of an abstraction [4].

3.5 A calendar bound from interval losses

The fully replanning opponent admits an additional public-boundary construction. Let $v_t(h)$ be the full-update continuation value at executed prefix h . For $t < e$, let $b_{t,e}(h)$ be the value when Blue commits over $[t, e)$, Red remains fully flexible, and the terminal payoff at e is v_e . Define

$$d(t, e) = \max_{h \text{ at } t} \{v_t(h) - b_{t,e}(h)\}. \quad (11)$$

Policy inclusion gives $d(t, e) \geq 0$, and $d(t, t+1) = 0$.

Theorem 8 (Calendar loss from public intervals). *For opponent F and boundaries $0 = t_0 < t_1 < \dots < t_m = T$,*

$$0 \leq V(F, F) - V(\{t_1, \dots, t_{m-1}\}, F) \leq \sum_{j=0}^{m-1} d(t_j, t_{j+1}). \quad (12)$$

Minimizing the right side over paths with at most $K+1$ edges from 0 to T gives a computable upper bound on the loss of its selected calendar.

Proof. At each selected boundary use a maximizing block policy of the game defining $b_{t,e}(h)$. Suppose the remaining stitched policy guarantees at least $v_e(h') - D$ at every successor prefix h' . Replacing v_e by that continuation lowers the guarantee by at most D . The selected block thus guarantees $b_{t,e}(h) - D \geq v_t(h) - d(t,e) - D$. Backward induction from zero terminal loss proves the sum. Every stitched decision uses an observed public prefix; Red has no private unexecuted block across these cuts. Finally each forward path is exactly one feasible public calendar, so shortest-path minimization preserves the bound. $\square$

With edge costs available, all budget bounds take $O(T^3)$ dynamic-programming work. Computing the full value tree and local interval games is preprocessing and can be expensive. Maxima over all histories can be conservative. The bound certifies sufficiency of a budget, not minimality for the true performance target. The construction is restricted to opponent F ; an unexecuted committed opponent suffix prevents the same public-boundary argument for C . The result specializes a minimax perturbation argument to calendar selection rather than claiming a new general abstraction theorem.

3.6 Correct use of symmetry

Exchange-equivariant dynamics and antisymmetric payoff imply $V(Es) = -V(s)$ after exchanging policy classes. At states fixed by the exchange, possibly combined with a verified spatial isometry, this implies zero value when policy classes are also exchanged into themselves. It does not imply that the finite-horizon value equals one-stage payoff at every state. For example, absorbing states $s = \pm 1$ with stage payoff s have two-stage value $2s$ while satisfying exchange oddness. Mixed minimax, rather than pure max-min, is also essential for the zero value of a skew-symmetric matrix. These distinctions govern the numerical symmetry checks and prevent an invalid degeneracy argument from entering the main line of reasoning.

4 Certified calendar optimization

4.1 An independent equilibrium reference

At own update t with next update e , the information set is indexed by (t, i_t, j_t) and has b^{e-t} block actions. The sum of their realization weights equals that of the parent block sequence. The parent is identified by the previous own update, its observed prefixes and the entire block that has now been executed. A unit root weight completes the flow equations. Terminal path pairs contribute their payoff to the corresponding pair of last block sequences. These realization constraints yield the standard perfect-recall sequence-form LP [7].

The implementation repairs proposed realization vectors top-down to nonnegative feasible behavioral distributions and reports the repair size. It then solves independent full best responses for both players. If p, q are the repaired strategies, the numerical certificate is based on

$$L = \min_{q' \in \mathcal{Q}_R} \mathbb{E}_{p,q'} A, \quad U = \max_{p' \in \mathcal{P}_S} \mathbb{E}_{p',q} A. \quad (13)$$

Residuals and numerical padding are recorded. These floating-point intervals do not include continuous-control discretization or physical model error and are not interval-arithmetic proofs.

4.2 Policy generation and common-boundary evaluation

Against a committed opponent, retain all b^T Red plans and a restricted set of Blue contingent policies. Solve the rectangular matrix game; its Blue mixture gives a valid full-opponent lower bound. Price a full Blue best response using Theorem 5 to obtain an upper bound and a new policy when needed. This follows the established response-oracle framework [1, 6].

In exact arithmetic a policy attaining a strictly positive response gap cannot already belong to the restricted set. Since the pure-policy space is finite, repeated addition terminates finitely. No polynomial iteration guarantee is asserted. A development pilot demonstrated that iteration counts vary substantially between physical instances, so the numerical portfolio fixes a 64-iteration generation budget and then falls back to the full sequence-form LP when necessary. Initial generation time is charged to the total cost. This rule was set before the frozen tests.

Against a fully flexible opponent, use the common updates of Theorem 3. Each local interval has committed Blue against flexible Red; exchanging the algebraic subgame’s roles applies the same one-sided solver. Distinct executed prefixes retain separate subgames. The intervals are propagated using Corollary 4.

4.3 Search on the calendar lattice

A branch-and-bound node specifies included updates I , excluded updates E and undecided updates. Every descendant is contained in $U = \mathcal{N} \setminus E$. Therefore an upper bound on $V(U, R)$ bounds the whole node, even when $|U| > K$. A feasible calendar supplies an incumbent lower bound L^* . A node is pruned only if its upper bound is at most $L^* + \varepsilon$. If $|I| > K$ it is infeasible. If $|I| = K$, only I remains admissible. If $|U| \leq K$, calendar U dominates its descendants by inclusion. Otherwise split on the smallest undecided update.

The basic method solves the relaxed calendar U . The enhanced method first tries the cheaper upper bound (9), using only opponent policies obtained from calendar queries already made by that method. A bound on U also bounds all its subsets. Neither variant assumes submodularity. Calendar solutions are reused across budgets within a method, and the complete budget-frontier cost is recorded.

Theorem 9 (Search certificate). *When every calendar node has been evaluated or safely pruned, the incumbent calendar has value at least L^* , and W_K is at most the largest final-node upper bound U^* . If $U^* - L^* \leq \varepsilon$, the returned calendar is ε -optimal.*

Proof. Binary inclusion/exclusion partitions all calendar subsets. Infeasible nodes contain no admissible solution. Both terminal reductions preserve an admissible dominating calendar by inclusion. Every remaining candidate therefore belongs to an evaluated or bounded terminal node. Their largest upper bound bounds every feasible value, while the incumbent lower bound is attained as a guarantee by a feasible strategy. Subtraction proves the claim. $\square$

This guarantee does not establish a universal speedup. A relaxation may be expensive, a transferred response bound may be loose, and an incumbent may coincide with a simple uniform calendar. These outcomes are retained in the computational comparisons.

4.4 Certifying the smallest useful budget

Given intervals $[L_K, U_K]$, $[L_0, U_0]$ and $[L_F, U_F]$, condition (6) is guaranteed when

$$L_K - \rho U_F - (1 - \rho)U_0 \geq 0, \quad (14)$$

and ruled out when

$$U_K - \rho L_F - (1 - \rho)L_0 < 0. \quad (15)$$

When the possible and guaranteed smallest budgets differ, both are reported. The numerical precision is never silently relaxed to force a unique budget.

Table 2: Predetermined computational design. Each physical configuration is evaluated separately against committed and flexible opponents. Shared baseline configurations are counted once in the 45-instance manifest.

Stage	Configuration
Development	Three geometries; focal speed ratios 0.85, 1, 1.15; distance16; $T = 6$, Grid-3
Frozen test	Three geometries; paired speed/distance conditions (.80, 13.5), (.95, 15), (1.05, 17), (1.20, 18.5); $T = 6$, Grid-3
Horizon	Three geometries; speed ratio1, distance16; $T = 4, 5, 6, 7$, Grid-3
Action grid	Same fixed anchors; $T = 4$, Grid-5 and Grid-7
Ablations	Same anchors at $T = 4$, Grid-3; one change at a time: angular averaging, terminal payoff, or rigid line-ahead formation

5 Computational design

5.1 Physical instances and controlled variations

The base instance contains three vessels per formation at arc-length offsets 0, 2, 4, a known initial geometry, constant operating speed, and bounded turn controls. The motion polyline has six substeps per epoch. The per-epoch interaction in (3) is evaluated at endpoints, with no damage updates or stochastic firing realization. The opponent speed is fixed at six model distance units per epoch; the focal speed is multiplied by the declared ratio. Both range ratios equal one. Speed is an imposed operating condition, not a nested set of selectable speed capabilities.

The directional kernel is the frozen CA-labeled fit. Its target-speed input remains fixed: a supplied value of six is clipped by the existing wrapper to five. Accordingly, a motion-speed comparison changes motion but does not represent a simultaneous recalibration of firing response. The inherited sector-boundary completion snaps angular differences within 10^{-8} degrees to the specified boundaries. It is applied to the physical scalar/vector evaluation, not to a projection of the payoff matrix onto a symmetric matrix.

Grid-3 has normalized controls $\{-1, 0, 1\}$ multiplied by a maximum turn of 60 degrees per epoch. Grid-5 adds $\pm 1/2$; Grid-7 uses increments of $1/3$. Both include Grid-3, but Grid-5 and Grid-7 are not nested in one another. The horizon remains fixed when calendars are compared.

The frozen tests provide 12 physical instances and 24 opponent conditions, with all 32 calendars at $T = 6$. Their speed/distance pairing is a computational test set, not a factorial experiment identifying the independent effect of speed. Discovery, implementation amendments and test results retain separate provenance. Older failed range and mobility interaction hypotheses are not reclassified as new evidence for the budget problem.

5.2 Controls, ablations and numerical precision

Every physical configuration has an independently generated player-swapped payoff matrix. Parallel geometry additionally requires reflection of signed turns. The designated speed-one development and first frozen-test anchors in each geometry also have independently solved mirror calendars. Algebraic role exchange inside a solver does not substitute for these physical checks.

Angular averaging integrates independent uniform shooter and target orientations. Shooter-sector weights are $1/6, 1/3, 1/3, 1/6$ and the target bow/stern weight is $1/3$. With identical kernels this isotropic net interaction cancels pairwise; it is an analytical control, not an empirical discovery. The terminal ablation uses $TL(h_T)$ to retain the integrated objective’s time scaling. The rigid line-ahead ablation rotates the whole formation with its leader instead of following the executed track.

The target numerical gap is $10^{-6} \max(1, \|A\|_\infty)$; response padding and propagated subgame error are recorded separately. Test cases include independent micro-game pure-policy enumer-

ation, arbitrary calendars, the legacy F/C endpoints, full-response reconstruction, information-boundary checks and non-submodular examples. Deterministic numerical intervals are not sampling confidence intervals. No population significance test is attached to a fixed grid of game instances.

5.3 Comparators and resource accounting

The comparators are complete calendar enumeration, basic and response-bound branch-and-bound, exact-increment greedy, and uniform, front-loaded and back-loaded calendars. The uniform calendar is $\{\lfloor jT/(K+1) \rfloor : j = 1, \dots, K\}$. All comparators hold physical input, budget and precision fixed. We report absolute and interval-valued regret, retained adaptation, full-space response gaps, oracle calls, runtime and peak memory, including non-improvements.

Two timing measurements are kept distinct. Online trace replay charges each requested calendar its individually recorded solver cost and executes response bounds; it measures computational accounting, not end-to-end wall-clock speedup. Cold runs recompute the prescribed development/test anchors with an empty method cache and a shared physical payoff input. Pay-off construction is listed separately. Calendar results are reused across budgets only within the same method.

The experiment uses one heavy local process on a 16GB machine, with an 8GB process-tree memory limit. Ordinary solver tasks are limited to 20minutes; bounded continuations use a cumulative six-hour instance budget. Resource limits, numerical stalls and unresolved values remain in the result record. No incomplete task is reported as exact convergence.

6 Results

All numbers below are read from the frozen solver outputs: 452 certified value files, of which 452 carry status `certified_numeric` and 0 are unresolved. The largest certified interval width over the whole set is $4.3e - 06$.

6.1 The budget frontier

On the development slice (three geometries, three focal operating speeds, $T = 6$, Grid-3, committed opponent), the full flexibility value ranges from +2.20 to +7.02, while the zero-revision value ranges from -0.67 to $+1.63$: the budget itself moves the value by up to 7.69 payoff units in this design. The frontier does *not* saturate early: the smallest budget attaining the full value is 5 in the cell where it is largest, and only 2 of 6 cells reach the full value with fewer than 5.0 revisions (median $K^* = 5$). Under this design the binding constraint is the budget itself, which is why the retention targets below are reported as certified budgets rather than as slack.

Table 3: Development slice, focal player against a committed opponent: the value without revisions, the full-flexibility value, and the smallest budget attaining it.

Geometry	focal speed	W_0	$W_{\max}$	K^*
head-on	0.85	+0.235	+6.816	5
head-on	1.00	+0.000	+5.348	5
head-on	1.15	+1.625	+7.015	4
parallel	0.85	-0.674	+2.200	4
parallel	1.00	+0.000	+3.407	5
parallel	1.15	+1.065	+5.276	5

6.2 Which revisions pay

For every budget the maximizing calendars concentrate on the early engaged epochs. Table ?? lists the selected calendar per cell; across the development slice the median first selected epoch is 1 and the median last selected epoch is 4. Because the calendar objective is not submodular (Proposition 2), this front-loading is an empirical property of the frozen instances, not a consequence of a greedy argument.

6.3 Loss certificates

The certificate experiment is reported once the frozen runs for this phase complete; the constructions themselves are Theorems 7 and 8, and a cell with an unresolved value is reported as an interval, not a ratio.

6.4 Minimum certified budget for a retention target

Table 4: Smallest certified budget retaining a fraction of the full-flexibility value, over the cells where the target is attainable with a certified value.

Retention	cells	median K
90%	9	3
95%	9	3
99%	9	4

In 5 condition(s) the full-flexibility value is indistinguishable from zero at the certified tolerance, so a *ratio* retention target is undefined. Those cells are reported as absolute losses, not as a minimum budget; this is the disclosure rule fixed before the runs.

6.5 Robustness and ablations

Section 5 fixes the horizon, action-grid and ablation protocols before their outputs. The corresponding summaries are reported in Figures ?? and ?? and in the supplement, with unresolved cells marked. We do not convert a change of sign under an ablation into a new positive claim; if an ablation overturns a pattern, that is reported as a boundary of the result.

7 Discussion and limitations

7.1 What the frontier does and does not settle

A certified frontier answers a budgeting question, not a doctrine question. It says which revision calendars are compatible with a tolerated loss under the frozen model, and it carries the residual of the computation that produced it. It does not say that a particular calendar is optimal in the real world, and it does not identify the mechanism by which revisions pay. Two readings of the same frontier should be resisted.

First, monotonicity in K is a correctness property, not a finding: it follows from policy inclusion (Proposition 1) and would hold for any model in this class. The informative content is the *shape*: where the frontier saturates, and which epochs the maximizing calendar selects.

Second, a budget that suffices is not a budget that is necessary. The deletion certificate bounds the loss of a specified coarse calendar from above; the interval certificate bounds a budget by a shortest path over interval losses. Neither claims minimality for a true performance target, and neither should be read as an optimality certificate for the calendar itself.

7.2 Where the certificates can be loose

Both certificates can be conservative, and we say where. The interval bound takes a maximum over all histories at a boundary ((11)), so one unfavourable history sets the cost of an interval for every path through it. The deletion certificate multiplies source transition kernels along hypothetical completions of the opponent’s future, which is exact for the constructed policy but says nothing about the best coarse policy. When either bound is far above the realised loss, the honest report is that the certificate is weak in that cell, not that the budget is tight. The certificates are also asymmetric in the opponent class: the interval construction needs public boundaries, which an unexecuted committed opponent suffix destroys, so it applies to the fully flexible opponent only.

7.3 Model scope

The numerical instantiation is a leader–follower line-ahead formation with fixed arc-length spacing, a frozen directional kernel calibrated against an auditable rules engine, and a fixed horizon with unit discount. Four restrictions bound the conclusions. (i) Formations are rigid in shape: no formation breaking, no independent vessel autonomy, and no command delay. (ii) Information is perfect within the model’s observation structure; the budget restricts *when* a player may act, not what it can see, so this paper is not about costly observation. (iii) The payoff is a reduced interaction kernel, not a physical ballistic or hydrodynamic model; it is a surrogate whose fidelity was measured separately and whose transfer to the rules engine is partial. (iv) The design is a fixed computational benchmark: conditions were declared before their outputs, the solver is deterministic, and no statistical sampling claim is attached to its output. Where a value is unresolved, we report the interval and the termination reason rather than a point estimate.

7.4 Falsified predecessor and its role

An earlier iteration of this project asked whether the side with the range advantage values revision rights differently from the side without it. Tested at this fidelity with exact anchors and frozen held-out configurations, that claim failed: most configurations pointed the other way, and neither the sign nor the ranking was stable. We keep the negative result in the record, do not re-tune the design to revive it, and do not restate it under a new name. The counterexample of Proposition 2 explains why the underlying intuition — that more revision opportunities must have diminishing value — cannot be assumed: revision opportunities can be complementary, so their value is a property of the instance.

7.5 What would move this to a stronger claim

Three things would strengthen the paper beyond its present scope, and we list them as open rather than claim them. A certificate that is tight rather than merely valid, for instance by computing interval losses on the realised history class instead of the worst history. A budgeted equilibrium notion in which both players choose calendars, since the present design fixes one calendar as public and optimises the other. And a formation model with deformation, which would test whether the frontier shape survives the representation change that the ablation probes only at $T = 4$.

8 Conclusion

We asked a budgeting question about adversarial trajectory planning: with revisions rationed to a public calendar, how much budget is needed, where should it be spent, and can a computation certify the answer? The paper’s answer has three parts.

The frontier is computable. For a history-dependent sealed-block game, the value attainable with at most K revisions can be computed exactly up to a certified interval, and the frontier is monotone by policy inclusion. On the frozen design the budget is not slack: the smallest budget attaining the full-flexibility value is the full budget in most cells, so the binding constraint is the number of revisions rather than the particular schedule. Where the budget does bite less, the maximizing calendars concentrate on the early engaged epochs — an empirical property, not a consequence of a greedy argument, because the calendar objective is not submodular.

The certificates are valid and their sharpness is reported. A constructive deletion certificate produces a feasible coarse policy together with an upper bound on its loss, and an interval-loss bound certifies a budget under a fully flexible opponent by a shortest path over public intervals. Both are bounds, not optimality claims: the interval bound maximises over histories and the deletion bound concerns one constructed policy, so we report where each is loose instead of presenting a bound as a tightness result.

The scope is bounded on purpose. The physical model is a leader–follower line-ahead formation with a frozen directional kernel; the design is a fixed computational benchmark rather than a statistical sample; and an earlier hypothesis that the range-advantaged side values revisions differently was falsified at this fidelity and is kept as a negative result. The contribution is the certified frontier and the structure that supports it, not a new general equilibrium-computation method.

The natural next questions are the ones the certificates cannot yet answer: tight rather than merely valid bounds, a game in which both sides select calendars, and a formation model that deforms.

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